RSHS

MATH O MAGIC

2·4

by

Karen Anderson

EDUCATIONAL DIRECTIONS PUBLICATIONS

INTRODUCTION

MATH O MAGIC 2·4 is part of a series of 12 books

MATHOMAGIC 1·2	MATHOMAGIC 4·3
MATHOMAGIC 1·3	MATHOMAGIC 4·4
MATHOMAGIC 2·3	MATHOMAGIC 5·3
MATHOMAGIC 2·4	MATHOMAGIC 5·4
MATHOMAGIC 3·3	MATHOMAGIC 6·3
MATHOMAGIC 3·4	MATHOMAGIC 6·4

These books are designed for students in years eight to ten.

1·2 and 1·3 are designed to revise the skills from earlier years and to be a platform from which high school skills are developed.

2·3, 3·3, 4·3, 5·3 and 6·3 develop skills in the SPACE, MEASUREMENT, and INFERENCE strands.

2·4, 3·4, 4·4, 5·4 and 6·4 address skills in the NUMBER and FUNCTION strands.

Stages 2, 3, 4, 5 and 6 indicate the level of difficulty, with each following in a sequential manner.

PREFACE

Often students find the necessary skill practice in Mathematics to be very boring. This series of books has been designed as a motivational tool to help the students to feel more enthusiastic about their work.

These worksheets are designed to be used in various ways –

1. MOTIVATION – Students are keen to solve the puzzle and keep asking for more.

2. INCENTIVE – As a reward for students who finish their work early.

3. SELF CHECKING
   – For students to be able to check their own work as they go along.

4. EXTRA PRACTICE
   – A fun way of doing extra practice on skills development.

5. HOMEWORK – an interesting way to get students to do more consolidation of mathematical skills in a way they enjoy.

6. REVISION – An exciting way to stimulate revision.

However you use these sheets, you will always be asked for more. The students enjoy working out the puzzles and riddles and get a great sense of satisfaction from getting them right.

Copies of this book may be obtained from:

EDUCATIONAL DIRECTIONS PUBLICATIONS
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THORNLIE 6108,
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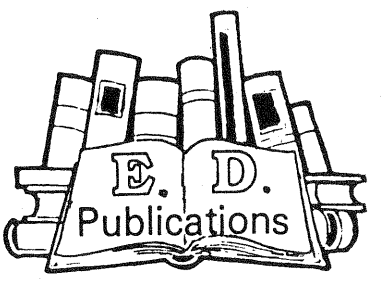
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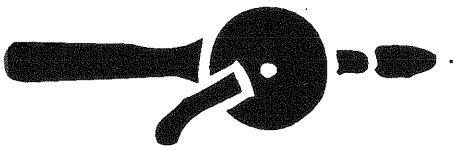
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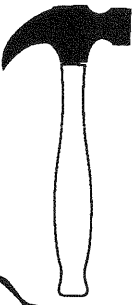
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WHAT DO YOU CALL A CARPENTER WHO MISPLACES HIS TOOLS?



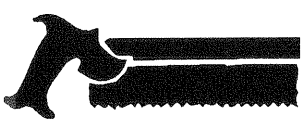
Simplify each of the following. Find the answer at the bottom of the page. Shade in the letter above each correct answer. When you finish you will be left with 9 letters. Write them in order from left to right to give the answer to the riddle.



8^2	900^2	100^2
20^2	$4\,000^2$	700^2
6^2	300^2	$5\,000^2$
$(0{\cdot}6)^2$	$(0{\cdot}3)^2$	$(0{\cdot}04)^2$
$(0{\cdot}5)^2$	$(0{\cdot}08)^2$	$(0{\cdot}02)^2$
$(\frac{1}{2})^2$	$(\frac{4}{9})^2$	$(\frac{9}{10})^2$
$(\frac{2}{5})^2$	$(\frac{1}{4})^2$	$(\frac{8}{9})^2$

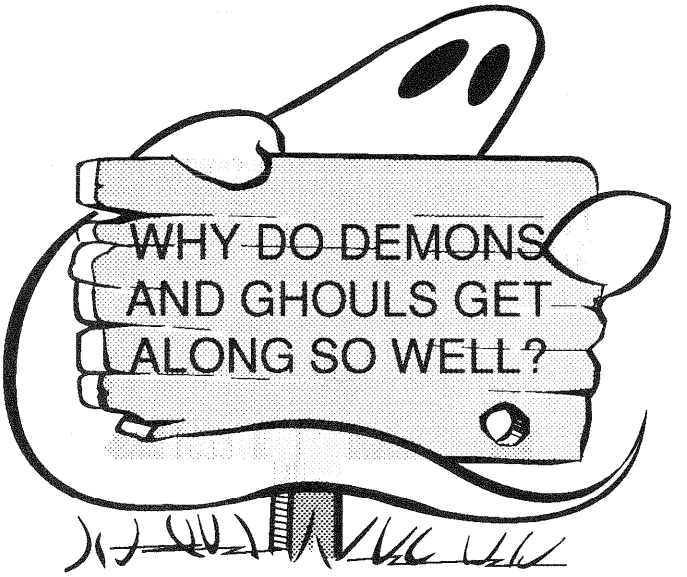


L	R	A	S	H	E	D	A	N	P	H	W	S	O	A
10 000	0·36	49 000	900	$\frac{1}{4}$	490 000	$\frac{1}{16}$	0·04	0·25	$\frac{4}{25}$	64	0·9	0·0004	$\frac{64}{81}$	16 000 000



N	L	K	M	O	N	T	A	S	E	A	E	N	R	S
$\frac{16}{81}$	0·64	810 000	0·09	0·6	90 000	$\frac{81}{100}$	0·0016	0·16	25 000 000	400	25 000	0·0064	4 900	36

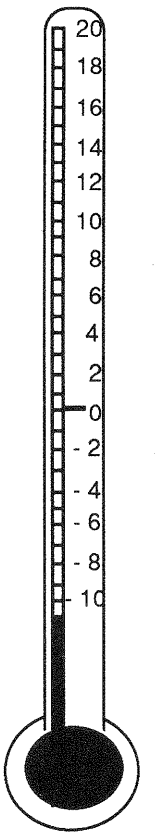
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Use this model of a thermometer to help you answer the following questions. Find the answer to each question in the box at the bottom of the page. Write the letter of each question in the correct box. When you finish you will have the answer to the riddle.

What is the temperature?

R 6° less than 9°C?
A 12° less than 4°C?
F 6° more than – 10°C?
N 12° more than – 5°C?
T 20° more than – 5° C?
C 10° less than 7° C?
O 15° less than 8° C?
H 8° less than 3° C?
I 5° more than – 7°C?
M 15° less than 6° C?
B 9° more than – 3° C?
D 10° more than – 2° C?
U 6° less than – 4° C?
E 4° more than – 10° C?
L 8° less than 7° C?
S 10° more than – 10° C?
G 8° more than – 3° C?



6°	– 6°	– 3°	– 8°	–10°	0°	– 6°	8°	– 6°	– 9°	– 7°	7°	0°	– 8°	3°	– 6°	– 8°
5°	– 5°	– 7°	-10°	– 1°		0°	6°	– 6°	0°	15°	– 4°	3°	– 2°	– 6°	7°	8°

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BOOKS FOR THE PRISON LIBRARY

"The Gangsters" by

$\overline{-8}$ $\overline{-6}$ $\overline{9}$ $\overline{6}$ $\overline{-9}$ $\overline{-10}$ $\overline{7}$ $\overline{-25}$ $\overline{-25}$ $\overline{-80}$ $\overline{-25}$

"My Life in Crime" by

$\overline{13}$ $\overline{-5}$ $\overline{7}$ $\overline{-6}$ $\overline{-9}$ $\overline{-6}$ $\overline{-3}$ $\overline{-6}$ $\overline{-6}$ $\overline{-2}$ $\overline{-25}$

"Bad Money" by

$\overline{32}$ $\overline{-6}$ $\overline{13}$ $\overline{-9}$ $\overline{7}$ $\overline{-8}$ $\overline{8}$ $\overline{6}$ $\overline{7}$ $\overline{16}$

"You Always Get Caught" by

$\overline{-10}$ $\overline{13}$ $\overline{-25}$ $\overline{-9}$ $\overline{-6}$ $\overline{-8}$ $\overline{-30}$ $\overline{-80}$ $\overline{-30}$ $\overline{7}$ $\overline{-25}$ $\overline{-8}$

To decode these authors, answer the questions below. Find your answers in the code. Each time the answer appears, write the letter of that question above it.

G If the temperature was 3° and it dropped 6°, what would the new temperature be?

R What number is the opposite of 8?

B What number is the opposite of – 9?

Z How many degrees are there between – 10°C and 6°C?

N During the day the temperature rose 20°. If it is now 11°C, what was the temperature earlier in the day before it rose?

A How could you use numbers to represent the situation "30 seconds before blast off"?

L If 80 m represents "80 metres above sea level", use numbers to represent "80 m below sea level".

C If it was – 12°C in Moscow and 20°C in Perth, how many degrees hotter is it in Perth than Moscow?

O If the temperature during the day was 4°C and it fell 10° during the night, what would the overnight temperature be?

S What number could be used to represent the time 10 seconds before a rocket is launched?

I London's overnight temperature was – 2°C. During the day it rose 8°. What was the maximum temperature for that day?

U How many degrees are there between – 3°C and 10°C?

P Overnight in Quebec the temperature was – 17°C. It rose 12° during the day. What was the maximum temperature for that day?

F How many degrees are there between – 10°C and – 2°C?

D What number is opposite to 2?

T What number is opposite to – 7?

E What number could be used to represent "25 m below sea level"?

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WHAT DO YOU GET IF YOU BLOW YOUR HAIR DRYER DOWN A RABBIT HOLE?

To answer this question, follow these instructions. In the left hand column are some number patterns. Draw a straight line connecting these patterns with the next 3 terms in each sequence. Each line will pass through a number and a letter. Write the letter in the correct box at the bottom of the page.

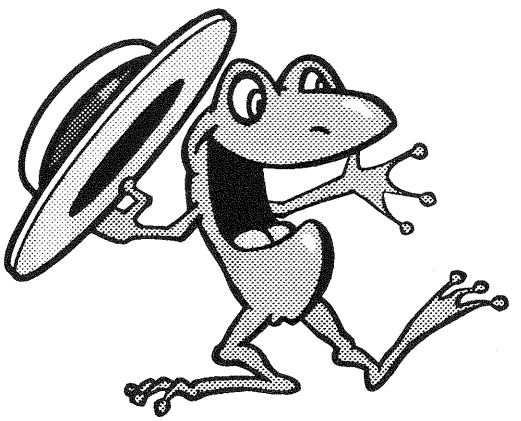
10, 8, 6, 4, . . .	*	1, – 1, – 3
13, 10, 7, 4 . . .	*	2, 0, – 2
20, 15, 10, 5, . . .	*	0, 3, 6
8, 5, 2, – 1, . . .	*	1, – 2, – 5
10, 6, 2, – 2, . . .	*	11, 16, 21
7, 5, 3, 1, . . .	*	– 1, – 3, – 5
11, 8, 5, 2, . . .	*	0, – 5, – 10
9, 7, 5, 3, . . .	*	4, 7, 10
11, 6, 1, – 4, . . .	*	– 9, – 14, – 19
7, 3, – 1, – 5, . . .	*	– 4, – 7, – 10
– 12, – 9, – 6, – 3, . . .	*	– 9, – 13, – 17
20, 10, 0, – 10, . .	*	– 3, – 1, 1
– 11, – 9, – 7, – 5, . . .	*	– 1, – 4, – 7
– 8, – 5, – 2, 1, . . .	*	– 20, – 30, – 40
– 9, – 4, 1, 6, . . .	*	– 6, – 10, – 14

1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*

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WHAT DO YOU GIVE SOMEONE WHO HAS EVERYTHING?

Decide whether the answers to the following will be **greater than** or **less than** the **first mentioned number**.
Circle the number–letter pair in the "greater than" or "less than" columns. Look for the same number–letter pair in the code below and cross it out. When you finish there will be 13 boxes remaining. Rearrange the letters according to numerical order to give the answer to the riddle.



	Greater than	Less than
25 x 3·9	2 – E	4 – S
0·8 x 0·11	7 – M	10 – T
16 x 2·75	9 – B	1 – P
43 x 1·89	6 – D	12 – S
17·2 x 0·82	13 – L	5 – T
0·91 x 0·45	8 – W	7 – E
11·4 x 2·4	11 – P	3 – N
18·3 x 0·11	4 – J	6 – M
62 x 1·7	10 – A	9 – T
41 x 0·65	5 – O	12 – T
15·3 x 2·4	1 – L	4 – A
39 x 0·5	12 – L	2 – H
65 x 1·5	6 – V	11 – H
24 x 0·75	7 – W	10 – K
43·6 x 0·4	9 – U	5 – H
95·1 x 0·98	13 – O	7 – U
0·3 x 2·8	3 – E	9 – Y

2 – B	9 – B	7 – A	10 – A	7 – U	4 – R
2 – H	3 – E	11 – A	5 – H	13 – M	12 – T
10 – L	1 – A	5 – T	5 – G	2 – E	9 – A
7 – E	10 – K	1 – L	11 – P	6 – V	6 – M
6 – L	6 – D	8 – R	12 – R	3 – U	10 – T

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WHEN THE JUDGE SAID: "*YOU HAVE A CHOICE: 30 DAYS OR 5 HUNDRED DOLLARS*," THE DEFENDENT REPLIED:



To complete this puzzle, decide whether the answers to the following will be ***greater than*** or ***less than*** the ***first mentioned number.*** Circle the number–letter pair in the "greater than" or "less than" columns. Look for the same number–letter pair in the code below and cross it out. When you finish there will be 15 boxes remaining. Rearrange the letters according to numerical order to give the answer to the riddle.

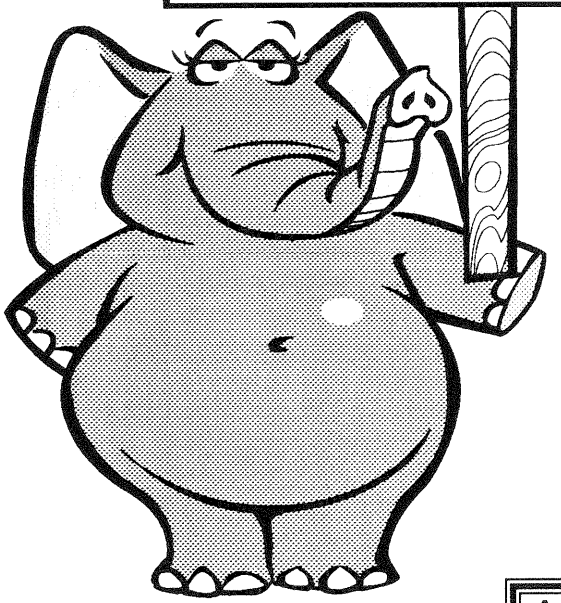
	Greater than	Less than
16·2 ÷ 3	**6 – M**	**5 – E**
47 ÷ 0·02	**9 – S**	**2 – P**
63 ÷ 17·1	**11 – O**	**9 – K**
8 ÷ 0·6	**2 – M**	**5 – R**
15 ÷ 1·3	**8 – P**	**1 – G**
22 ÷ 4·8	**14 – O**	**15 – J**
33·1 ÷ 0·63	**7 – U**	**4 – V**
14·8 ÷ 0·95	**10 – F**	**12 – D**
28·2 ÷ 2·2	**3 – R**	**4 – B**
85 ÷ 3·6	**9 – S**	**10 – A**
25·3 ÷ 0·5	**15 – R**	**6 – C**
82·4 ÷ 2·5	**1 – N**	**2 – T**
68·4 ÷ 6·2	**7 – D**	**10 – U**
75·8 ÷ 0·25	**8 – S**	**13 – Y**
95 ÷ 0·75	**5 – P**	**14 – R**

6 – K	15 – J	9 – S	4 – T	4 – B	2 – L
7 – U	14 – E	1 – G	10 – E	15 – Y	8 – S
11 – M	3 – L	10 – F	13 – N	2 – T	9 – H
15 – R	5 – E	10 – U	10 – A	7 – E	2 – M
12 – O	5 – P	5 – A	1 – I'	8 – T	9 – K

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What do you get when you cross a parrot with a woodpecker?

Answer each of the following questions. Look for your answers in the code below. Wherever you see the answer, write the letter corresponding to the question above it.



A large bucket contains 24 litres of water. How many jugs of water could be filled from the bucket if the jugs had capacities of:

- **M** 4 litres?
- **U** 12 litres?
- **D** 0·6 litres?
- **O** 0·5 litres?

A factory worker earns $10·40 per hour. How many dollars would they earn if they worked for:

- **C** 8 hours?
- **T** 40 hours?
- **I** 0·5 hours?
- **K** 15 minutes?

A plank of wood 16 metres long is to be cut into pieces. How many pieces could be cut if they had lengths of:

- **A** 2 metres?
- **S** 4 metres
- **R** 0·5 metres?
- **Y** 10 centimetres?

Apples are selling for $2·50 per kilogram.
How many dollars would you expect to pay if you bought:

- **N** 4 kg?
- **H** 10 kg?
- **L** 0·5 kg?
- **E** 0·25 kg?

8	10	8	10	5·20	6	8	1·25	416	25	8	416

416	8	1·25	2·60	4	416	48	160	48	2	5·20	10

6	48	32	4	0·62	83·20	48	40	0·62

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WHY DID THE BURGLAR TAKE A BATH BEFORE BREAKING OUT OF JAIL?

Instructions: *Convert each of the fractions below to decimals. Look for each decimal equivalent in the Code Key and notice the letter next to it. Write the letter for that question in the appropriate box at the bottom of the page. Continue until you have the answer to the riddle.*

1. $\frac{1}{2}$
2. $\frac{5}{16}$
3. $\frac{5}{8}$
4. $\frac{1}{5}$
5. $\frac{4}{5}$
6. $\frac{17}{20}$
7. $\frac{1}{4}$
8. $\frac{19}{20}$
9. $\frac{7}{16}$
10. $\frac{1}{8}$
11. $\frac{7}{8}$
12. $\frac{21}{25}$
13. $\frac{2}{5}$
14. $\frac{3}{10}$
15. $\frac{3}{4}$
16. $\frac{3}{5}$
17. $\frac{9}{10}$
18. $\frac{7}{10}$
19. $\frac{1}{3}$

CODE KEY

E	0·125
G	0·2
N	0·25
E	0·3
C	$0\dot{3}$
L	0·3125
A	0·4
O	0·4375
A	0·5
Y	0·6
M	0·625
T	0·7
E	0·75
T	0·8
W	0·84
A	0·85
A	0·875
K	0·9
A	0·95

☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
5	9	3	13	17	10	1	19	2	15	11	7

☐	☐	☐	☐	☐	☐	☐
4	14	18	8	12	6	16

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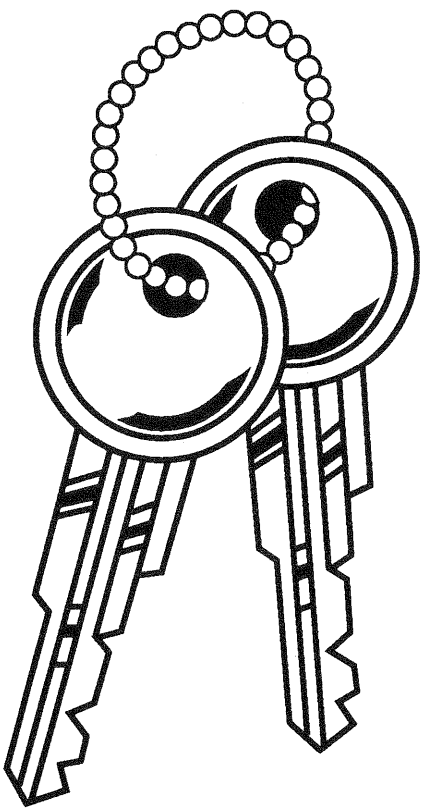
WHAT DID THE LOCKSMITH DO WHEN HIS SHOP CAUGHT ON FIRE?

To answer this question, ***MENTALLY*** *determine each of the following amounts. Do* ***NOT*** *use your calculator. Find your answer in the Code Key and notice the letter next to it. Write the letter above the question number at the bottom of the page.*

1.	10% of $540
2.	10% of $720
3.	20% of $720
4.	50% of $720
5.	200% of $720
6.	$12\frac{1}{2}$% of $720
7.	$33\frac{1}{3}$% of $720
8.	$66\frac{2}{3}$% of $720
9.	20% of $65
10.	40% of $65
11.	60% of $65

CODE KEY

H	$39
B	$360
R	$90
E	$480
M	$72
A	$26
T	$1 440
F	$240
O	$13
D	$54
L	$144



___	___	___	___	___	___	___	___	___	___	___
11	8	2	10	1	8	10	4	9	3	5

___	___	___	___	___	___	___	___	___	___
7	9	6	5	11	8	1	9	9	6

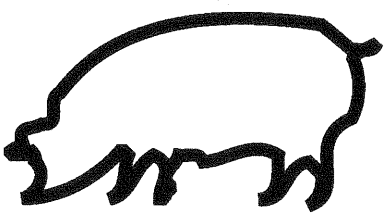
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WHAT DO YOU GET WHEN YOU CROSS A PIG WITH A CENTIPEDE?

MENTALLY *determine each of the following amounts. Find your answer in the right hand column. Rule a line connecting the question and your answer. Each line will pass through a number and a letter. Write the letter in the correct box at the bottom of the page, and you will have the answer to the riddle.*

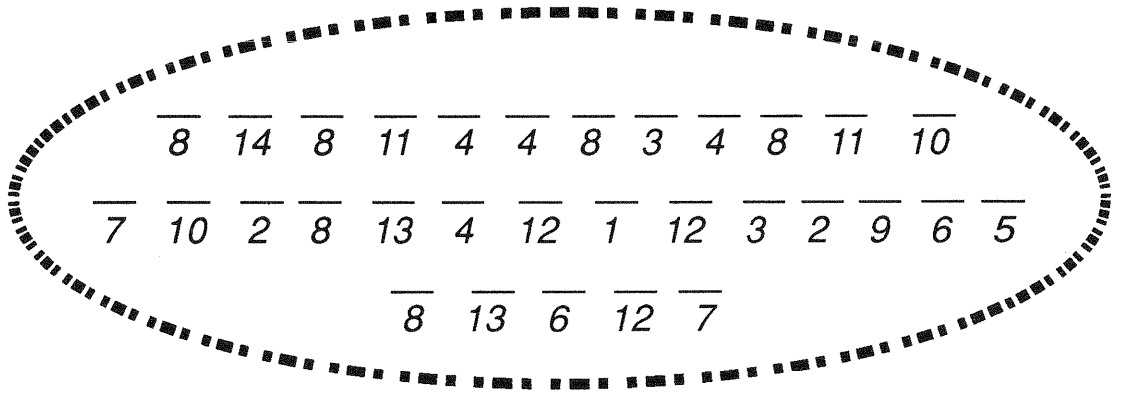
10% of $780	*	$55
90% of $780	*	$230
5% of $780	*	$78
95% of $780	*	$495
33 $\frac{1}{3}$% of $690	*	$741
66 $\frac{2}{3}$% of $690	*	$165
10% of $690	*	$69
90% of $690	*	$702
10% of $550	*	$330
30% of $550	*	$460
60% of $550	*	$621
90% of $550	*	$39

5	11	7	9	3	12	4	8	1	10	2	6



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SIGN IN FRONT OF A CLOCK REPAIR SHOP:



Instructions:

Solve each of the following problems. Find your answer in the Code Key and notice the letter next to it. Write that letter above the number of the question in the spaces above. Keep working until you have solved the puzzle.

1. Find 68% of 350.
2. Find 24% of 240.
3. Find 55% of 640.
4. Find 78% of 680.
5. A school has a population of 525 students. Of these, 48% are girls. How many are girls?
6. A company produces 1 600 pieces of confectionery each hour. 23% are chocolates. How many chocolates are produced each hour?
7. A person earns $22 500 each year. They pay 33% in tax. How much tax do they pay?
8. A stereo is priced at $990. During a sale it is advertised with a 17% discount. How much is the discount?
9. How much would a person pay for the stereo in question 8?
10. Last year, Susan was 150 cm tall. This year her height grew by 12%. By how much did she grow?
11. How tall is Susan now?
12. A shop had $18 400 worth of stock. During a fire, 22% of the stock was damaged. What value of stock was damaged?
13. On a test, John scored 90%. If the test was out of 40, how many marks did John score?
14. In her science test, Janet scored 80%. If the test was out of 90, how many marks did Janet score?

CODE KEY

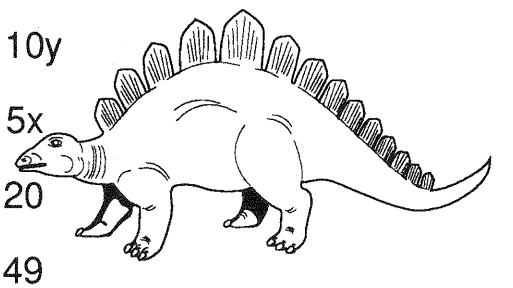
S	18 cm
Z	$821·70
U	72
O	530·4
E	368
N	238
D	252
H	36
Y	57·6
A	$4 048
P	$7 425
L	352
K	168 cm
C	$168·30

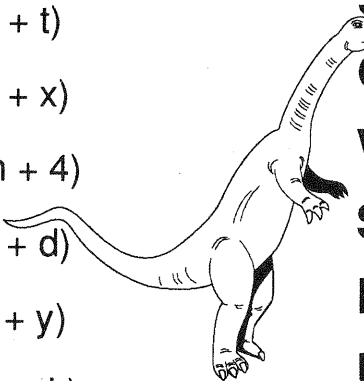
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WHAT DO YOU GET IF TWO PREHISTORIC MONSTERS CRASH INTO EACH OTHER AT 200 KM PER HOUR?

To answer this question, match each product below with its simplified form. Notice the letter for each question. Write that letter above the number at the bottom of the page.

1.	4(x + 2)	**E**	6a + 6d
2.	5(y + 4)	**T**	12 + 6m
3.	6(2 + m)	**N**	16 + 4m
4.	10(a + 4)	**N**	8f + 16g
5.	5(5 + k)	**R**	4x + 8
6.	7(f + 5)	**U**	9y + 81
7.	8(x + 2)	**R**	7f + 35
8.	3(9 + y)	**A**	5x + 10y
9.	4(4 + m)	**A**	30 + 5x
10.	7(x + 7)	**R**	5y + 20
11.	9(y + 9)	**Y**	7x + 49
12.	4(6 + t)	**S**	10a + 40
13.	5(6 + x)	**O**	8x + 8y
14.	7(m + 4)	**W**	27 + 3y
15.	6(a + d)	**S**	8x + 16
16.	8(x + y)	**K**	7m + 7k
17.	7(m + k)	**U**	7m + 28
18.	5(x + 2y)	**S**	25 + 5k
19.	8(f + 2g)	**C**	24 + 4t



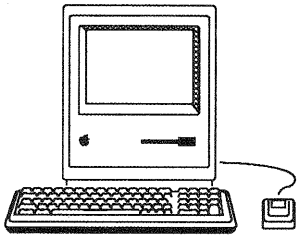


3	10	6	13	19	9	16	4	18	11	2	14	7

8	1	15	12	17	5

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WHAT DO YOU GET WHEN YOU CROSS A COMPUTER PROGRAMMER WITH AN OLYMPIC ATHLETE?

To answer this question, match each product below with its simplified form. Notice the letter for each question. Write that letter above the number at the bottom of the page.

1.	3(p + 4)	**O**	6p + 18	**6.**	8(p + 4)	**P**	11p + 77
2.	2(p + 5)	**H**	7p + 49	**7.**	5(p + 2)	**C**	8p + 32
3.	7(p + 7)	**S**	2p + 10	**8.**	10(5 + p)	**U**	5p + 15
4.	(p + 3)6	**R**	7p + 35	**9.**	(p + 3)5	**T**	50 + 10p
5.	(p + 5)7	**D**	3p + 12	**10.**	11(p + 7)	**A**	5p + 10
11.	7(4 + p)	**P**	6p + 24	**16.**	8(p + 3)	**Y**	10p + 80
12.	(5 + p)9	**F**	45 + 9p	**17.**	10(p + 8)	**L**	5p + 15
13.	(4 + p)5	**E**	28 + 7p	**18.**	2(p + 7)	**I**	2p + 14
14.	4(p + 5)	**S**	4p + 20	**19.**	(p + 3)5	**W**	18 + 3p
15.	6(p + 4)	**R**	20 + 5p	**20.**	3(6 + p)	**O**	8p + 24

7	12	19	4	10	15	17	1	18	14	6	9	2	8	3	13	16	20	11	5

What game do MONSTER children play?

To answer this question, follow these instructions. Match a binomial in the left hand column with its factored form in the right hand column. Draw a straight line connecting each pair. Each line will pass through a number and a letter. Write the letter in the correct box at the bottom of the page.

3x + 9y	*	7(k + 5)
8a + 4b	*	9(t + 4)
10m + 30k	*	3(x + 3y)
15s + 5t	*	5(3s + t)
16x + 8y	*	8(3 + w)
12a + 4b	*	4(2a + b)
4x + 32	*	4(x + 8)
21 + 7m	*	2(d + e)
7k + 35	*	10(m + 3k)
2d + 2e	*	10(x + y)
10x + 10y	*	4(3a + b)
24 + 8w	*	7(3 + m)
9t + 36	*	8(2x + y)

2	10	5	8	13	4	11	6	1	9	7	12	3

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DID YOU HEAR ABOUT THE DRACULA DOLL? YOU PUT IN A BATTERY AND IT . . .

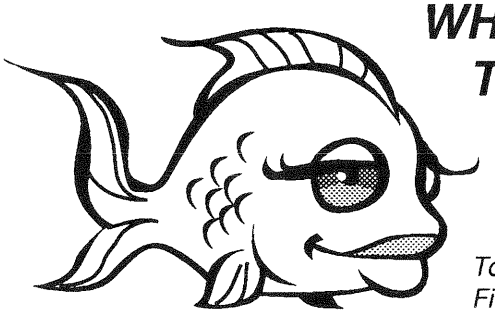
INSTRUCTIONS: *Factorize each binomial below. Notice the letter next to each factored form in the right hand column. Write the letter above the number for each question at the bottom of the page.*

1.	12w + 18	**E**	5(4 + e)
2.	10w + 15e	**O**	5(3w + 5e)
3.	21w + 28e	**I**	7 (w + e)
4.	30w + 18e	**L**	3(w + 2)
5.	15w + 25e	**S**	6(2w + 3)
6.	18w + 24e	**D**	7(3 + e)
7.	4w + 4e	**C**	4(w + 4)
8.	3w + 3e	**B**	5(2w + 3e)
9.	2w + 6	**T**	4(3 + e)
10.	20 + 5e	**K**	6(3w + 4e)
11.	9w + 18	**B**	6(5w + 3e)
12.	10w + 30	**N**	3(2 + e)
13.	21 + 7e	**R**	3(10 + e)
14.	4w + 20	**H**	3(w + e)
15.	2w + 2e	**A**	5(w + e)
16.	3w + 6	**N**	4(w + 5)
17.	4w + 16	**E**	6(w + 4)
18.	5w + 5e	**I**	2(w + 3)
19.	6 + 3e	**I**	10(w + 3)
20.	5w + 30	**A**	2(w + e)
21.	12 + 4e	**B**	9(w + 2)
22.	16 + 4e	**L**	4(4 + e)
23.	30 + 3e	**E**	4(w + e)
24.	6w + 24	**T**	5(w + 6)
25.	7w + 7e	**E**	7(3w + 4e)

—	—	—	—	—	—	—	—	—	—	—	—	—
4	12	20	7	1	18	11	15	23	2	9	3	
—	—	—	—	—	—	—	—	—	—	—	—	—
13	5	16	22	25	14	21	8	24	19	10	17	6



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WHAT SITS ON THE BOTTOM OF THE SEA AND MAKES YOU AN OFFER YOU CAN'T REFUSE?

To answer this question, factorize each binomial below. Find its factors in the boxes below. Cross out each box containing a correct answer. When you finish, write the remaining letters in order from left to right.

5x + 15	12d + 6	6x + 6y
4f + 4g	7y + 28	11p + 11q
8t + 32	49 + 7w	45 + 9g
10a + 10b	6 + 6x	20 + 15y
50m + 30	6x + 15	24 + 12k
35 + 28z	40 + 12d	12a + 8f
10x + 15y	32a + 24d	4a + 6
10x + 15	9 + 6y	8a + 12b

6(1 + x) AC	4(2a +3b) ED	6(x + y) PA	4(6 + 5x) TH	7(5 + 4z) FE	5(x + 3) ST
2(a + m) EC	3(3 + 2y) NE	6(2d + 1) ND	2(2a + 3) ES	10(a + b) TH	11(p + q) ED
3(2x + 5) YS	4(f + g) DA	4(3a + 2f) PS	12(2 + k) EA	7(7 + w) GU	3(3m + 7) OD
5(2x + 3) PH	5(a + b) FA	8(4a + 3d) ST	10(5m + 3) OA	7(y + 4) ON	9(5 + g) HA
5(2x + 3y) ND	4(2 + 3x) TH	4(10 + 3d) IN	5(4 + 3y) CE	3(2d + 5) ER	8(t + 4) RS

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FISH LOVER'S LIBRARY

"Waterways of the World" by

4	7	2		4		5	10	9	10	1

"A Fish Story" by

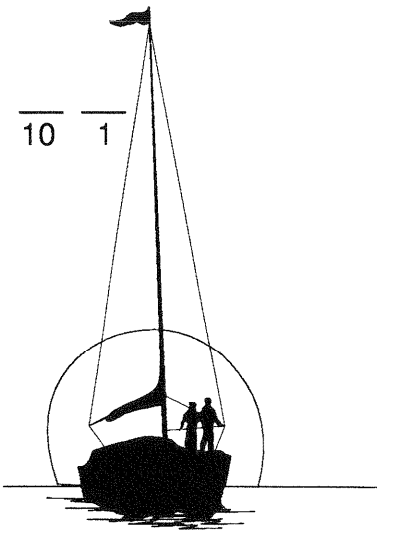
5	14	10	15		8	2	10	9

"Caught in the Flood" by

9	11	10	6		14	10	15	13

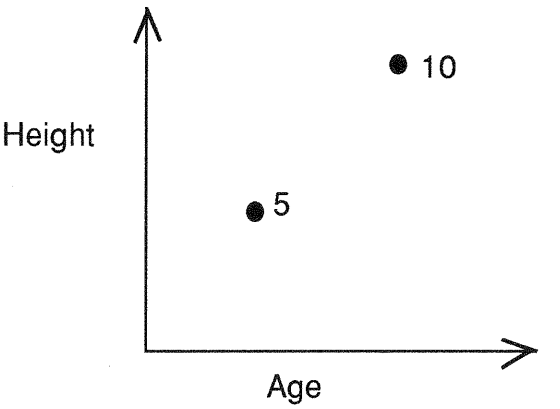
"Little Fishes" by

10	9	9	2		5	6	11	12	2	3

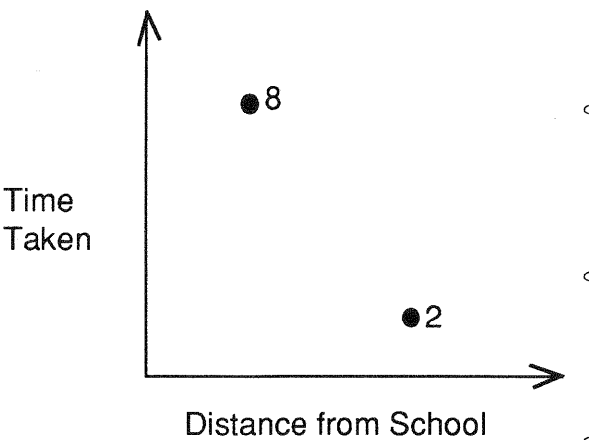


Read each group of statements below. Write the correct letters next to the points on each graph. Notice the number next to each letter. Write the letter above the matching number in the puzzle above.

A is older than C.
C is shorter than A.



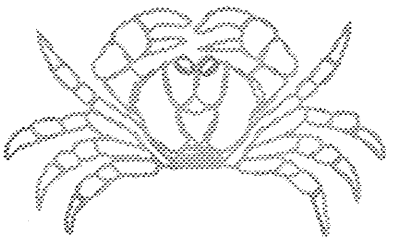
D lives closer to school than E.
D takes longer to travel to school than E.

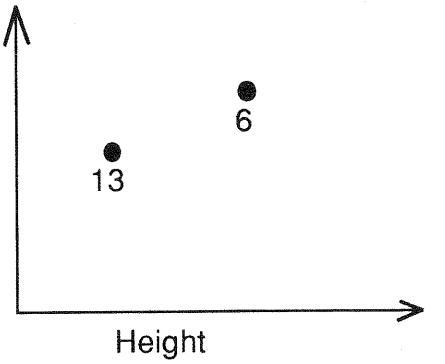


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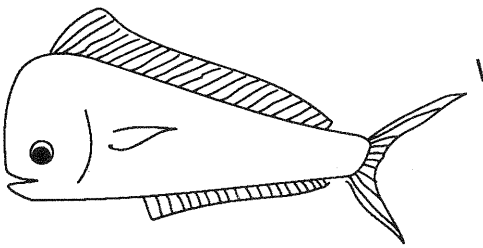
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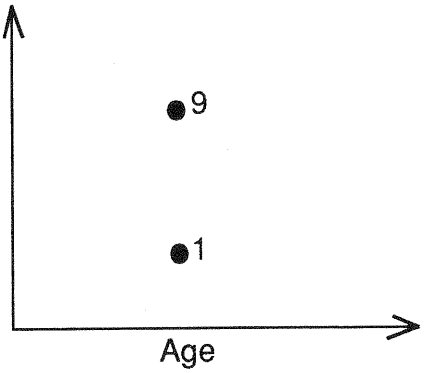
Flask H is taller than flask K.
Flask H holds more water than flask K.



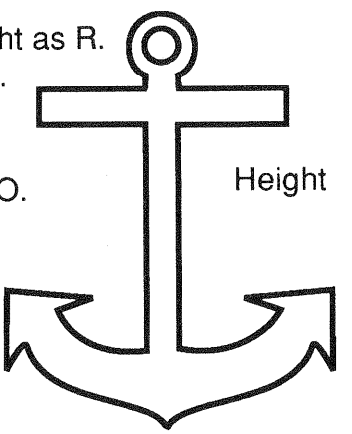


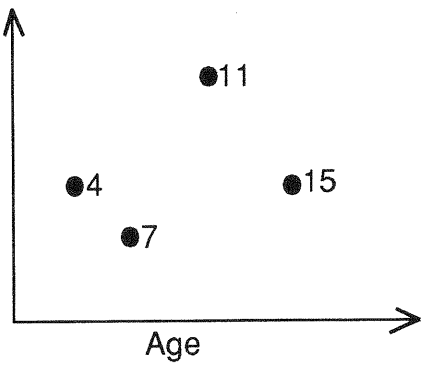
L is lighter than N.
L and N are the same age.



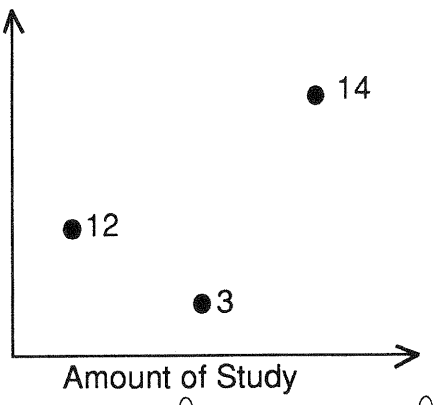


O is taller than R.
S is the same height as R.
U is shorter than S.
R is older than O.
S is the youngest.
U is younger than O.





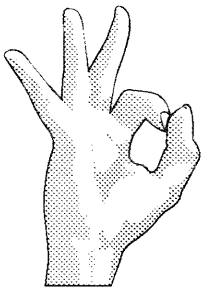
Z did a large amount of study and received a high test score.
V did a very low amount of study and received a mediocre test score.
Y did a fairly large amount of study and received a very low test score.



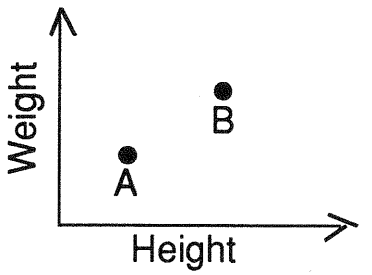
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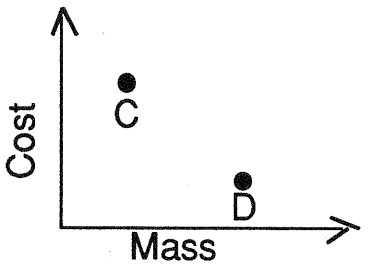
SIGN IN FRONT OF A MAGIC SHOP:

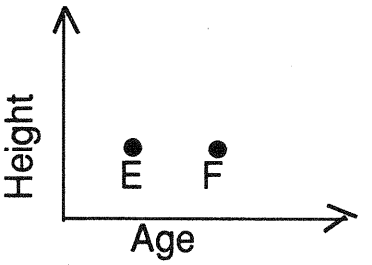
4	11	16	8	1	5	15	19	3	9	14
	6	18	10	7	12	17	2	13		

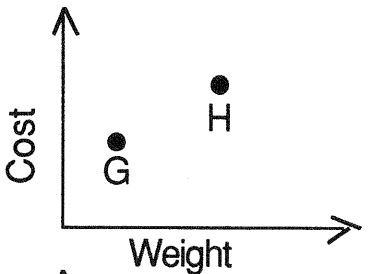


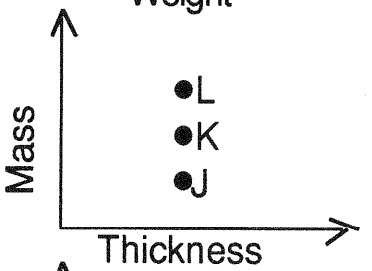
To decode this puzzle, decide whether each statement below is true or false. For each question, circle a letter in the true/false column. Notice which letter is circled for each number question and write it in the correct space in the answer key above.

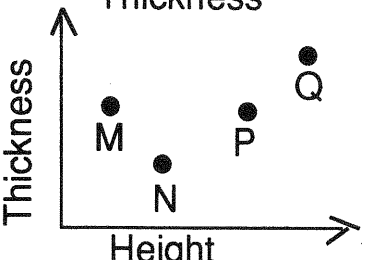












		TRUE	FALSE
1.	A is shorter than B.	P	S
2.	A is lighter than B.	C	N
3.	C is heavier than D.	A	R
4.	D is cheaper than C.	D	E
5.	E is older than F.	S	P
6.	F is taller than E.	G	F
7.	E and F are the same age.	E	L
8.	H is more expensive than G.	A	N
9.	H is heavier than G.	E	O
10.	H is taller than G.	L	R
11.	L is thicker than J.	E	I
12.	L is heavier than J.	U	A
13.	K is lighter than L.	H	E
14.	K is thicker than J.	N	D
15.	J and K have the same mass.	S	E
16.	Q is thicker than M.	S	T
17.	P is thicker than N.	N	D
18.	N is shorter than M.	S	O
19.	M and P are the same height.	E	A

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WHAT HAPPENED TO THE FARMER WHO FELL THE WELL?



To answer this question, solve each problem below. Find the answer to each problem in the code. Write the letter of each question above the number answer. Keep working until you have the answer to the riddle.

H Seven more than a number is nineteen. What is the number?

U Nine less than a number is eight. What is the number?

D Six more than twice a number is twenty. What is the number?

B Seven more than three times a number is thirty-one. What is the number.

E Ten less than twice a number is twelve. What is the number?

T Eight less than five times a number is twenty-two. What is the number?

I One third of a number is eight. What is the number?

C One fifth of a number is six. What is the number?

K Two thirds of a number is six. What is the number?

12 11 9 24 30 9 11 7 6 12 11

8 17 30 9 11 6

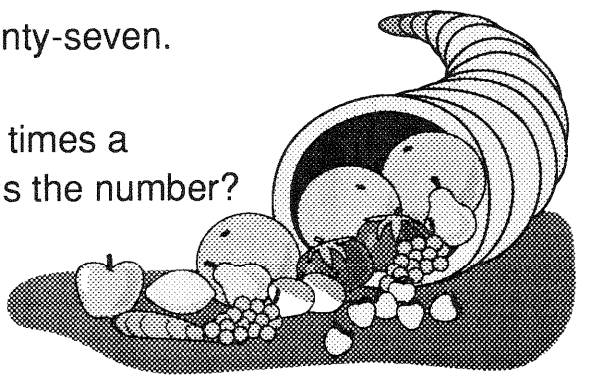
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WHAT SALAD DO PEOPLE PREFER WHEN THEY WANT PRIVACY?

Solve each word problem below. Find your answer in the Code Key. Notice the letter next to it. Write the letter for that question above the number of the question in the spaces at the bottom of the page.

1. When a number is multiplied by three and four is added to it, the result is twenty-two. What is the number?
2. Two more than the square of a number is twenty-seven. What is the number?
3. When seven is added to a number and then multiplied by three, the result is thirty-three. What is the number?
4. Six more than the square of a number is seventy. What is the number?
5. When three is subtracted from a number and then multiplied by six, the result is twenty-four. What is the number?
6. The square root of a number is seven. What is the number?
7. Two less than the square root of a number is six. What is the number?
8. Four more than the square root of a number is thirteen. What is the number?
9. When a number is divided by five and then four is subtracted, the result is two. What is the number?
10. Seven more than twice a number is twenty-five. What is the number?
11. Three quarters of a number is twenty-seven. What is the number?
12. When six is subtracted from three times a number, the result is thirty. What is the number?

CODE KEY	
49	L
6	E
9	O
7	E
81	A
36	T
8	E
12	C
5	N
64	U
30	T
4	L



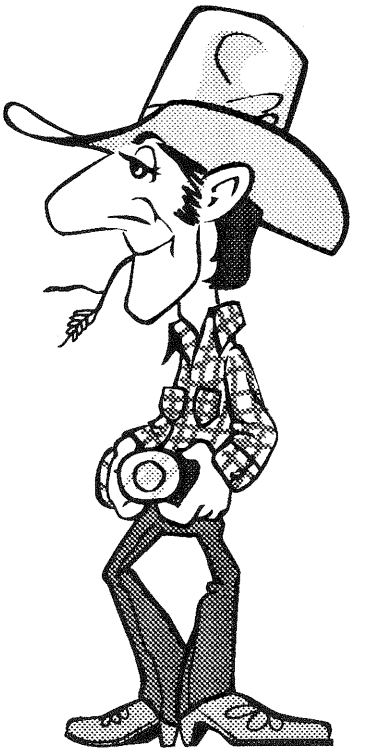
—	—	—	—	—	—	—	—	—	—	—	—
6	1	9	11	7	12	5	8	3	10	2	4

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Why was the farmer arrested in the morning?

Find the value of each variable below. Find your answer in the spaces at the bottom of the page. Write the letter of each equation above the appropriate number.

H	16 − h = 5
S	s − 2 = 10
A	a + 12 = 17
F	11 − f = 2
Y	y − 9 = 20
C	c − 8 = 12
U	5 − u = 1
B	16 − b = 2
G	g + 12 = 30
T	22 − t = 7
O	20 + 0 = 28
E	16 − e = 3
N	n − 12 = 15
R	14 − r = 8
I	i + 16 = 26



$\overline{14}$ $\overline{13}$ $\overline{20}$ $\overline{5}$ $\overline{4}$ $\overline{12}$ $\overline{13}$ $\overline{11}$ $\overline{13}$ $\overline{11}$ $\overline{10}$ $\overline{15}$

$\overline{15}$ $\overline{11}$ $\overline{13}$ $\overline{11}$ $\overline{5}$ $\overline{29}$ $\overline{15}$ $\overline{11}$ $\overline{13}$

$\overline{27}$ $\overline{10}$ $\overline{18}$ $\overline{11}$ $\overline{15}$ $\overline{14}$ $\overline{13}$ $\overline{9}$ $\overline{8}$ $\overline{6}$ $\overline{13}$

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WHAT DO YOU GET IF YOU CROSS A STICK OF DYNAMITE WITH THE WHITE OF AN EGG?

Solve each equation below. Find your answer in the Code Key. Notice the letter next to it. Write that letter above the number of the question in the spaces at the bottom of the page.

1. $x - 19 = 30$
2. $30 - y = 15$
3. $k - 17 = 20$
4. $m + 11 = 33$
5. $j + 18 = 36$
6. $v - 10 = 19$
7. $t - 25 = 40$
8. $w - 12 = 15$
9. $r + 17 = 34$
10. $40 - k = 15$
11. $9 + k = 35$
12. $12 + y = 50$
13. $x + 16 = 40$

CODE KEY	
I	25
B	17
N	15
E	18
O	27
A	29
G	65
E	37
R	24
O	26
M	38
U	49
M	22



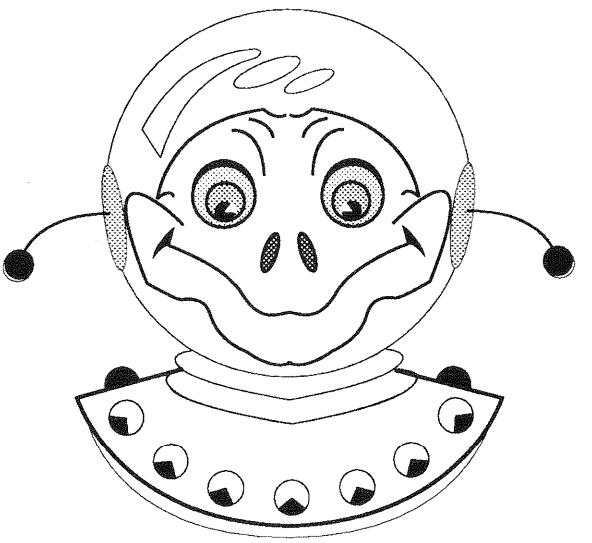
6	9	11	8	4		12	5	13	10	2	7	1	3

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HOW DO MARTIANS SHAVE?

To answer this question, solve each equation below. Find the solution in the right hand column. Rule a line to connect the equation with the solution. Each line will pass through a letter and a number. Write the letter in the correct space at the bottom of the page.

a + 7 = 13	*	* 18
12 + b = 16	*	* 2
c − 3 = 10	*	* 4
10 − d = 5	*	* 15
e + 7 = 19	*	* 10
19 + f = 21	*	* 13
g + 4 = 18	*	* 9
h − 6 = 9	*	* 16
i − 4 = 3	*	* 6
13 − j = 4	*	* 3
k − 8 = 8	*	* 11
12 + l = 23	*	* 12
m + 10 = 20	*	* 14
19 − n = 1	*	* 5
p + 9 = 12	*	* 7



5	7	10	6	12	15	11	3	14	8	1	4	13	2	9

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WHAT KIND OF CASE WOULD A LAWYER HAVE IF HE SLIPPED AND HURT HIMSELF AT THE POOL?

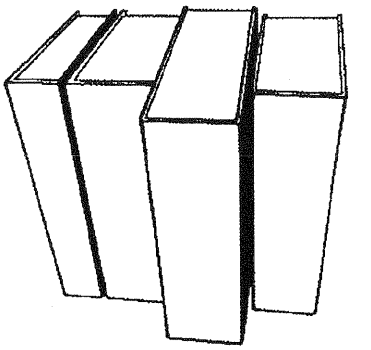
Solve each equation below. Find your answer in the right hand column. Rule a line to connect the equation with its solution. Each line will pass through a letter and a number. Write the letter in the box above the appropriate number.

$12 + 4b = 32$	*	4
$7j - 10 = 60$	*	50
$10k - 7 = 33$	*	10
$200 - 3w = 56$	*	15
$6x + 10 = 310$	*	25
$2a + 10 = 64$	*	33
$12 + 3d = 90$	*	27
$100 - 2m = 34$	*	18
$120 - 6b = 12$	*	5
$9x + 15 = 150$	*	12
$3w + 14 = 50$	*	26
$5x - 30 = 95$	*	48

1	5	9	12	10	4	11	7	2	8	6	3

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IF TWO AUTHORS SUED EACH OTHER, WHAT WOULD YOU HAVE?



Solve each equation below. Find the answer in the boxes at the bottom of the page. Cross out each box containing a correct answer. When you have finished, write the remaining nine letters in order from left to right.

2a + 8 = 14	5b + 4 = 69	5 + 4c = 69
2d + 7 = 51	6 + 3e = 81	2f + 10 = 210
200 − 3x = 71	100 − 5b = 25	4c − 7 = 137
3x + 3 = 60	12 − 2d = 10	100 − 5m = 0
10g + 8 = 308	11 + 4h = 179	6f − 3 = 39
21 − 2k = 3	25 − 3d = 1	16 + 2t = 16
30 − 4f = 6	24 + 3j = 30	10 + 3k = 46

N 15	T 19	R 13	A 5	E 9	S 42	B 18	D 6	O 27	M 22
T 100	O 48	R 2	K 50	S 43	P 8	N 3	O 36	E 7	C 4
A 26	D 1	M 12	A 16	C 0	S 33	P 20	E 29	T 25	S 30

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MATH O MAGIC 2·4 SOLUTIONS

1. A SAW LOSER
2. BECAUSE DEMONS ARE A GIRL'S BEST FRIEND
3. ROBIN STEELE
   UPTON O. GOODE
   COUNT R. FITZ
   SUE NORA LATER
4. HOT CROSS BUNNIES
5. A BURGLAR ALARM
6. I'LL TAKE THE MONEY
7. AN ANIMAL THAT TALKS TO YOU IN MORSE CODE
8. TO MAKE A CLEAN GETAWAY
9. HE MADE A BOLT FOR THE DOOR
10. BACON AND LEGS
11. CUCKOO CLOCKS PSYCHOANALYZED CHEAP
12. TYRANNOSAURUS WRECKS
13. A FLOPPY DISCUS THROWER
14. HYDE AND SHRIEK
15. BITES BARBIE DOLL IN THE NECK
16. THE CODFATHER
17. SUE S. CANAL
    CZAR DEAN
    NOAH ZARK
    ANNE CHOVEY
18. DISAPPEARED FOR LUNCH
19. HE KICKED THE BUCKET
20. LETTUCE ALONE
21. BECAUSE HE HIT THE HAY THE NIGHT BEFORE
22. A BOOM – MERINGUE
23. WITH LASER BLADES
24. A BATHING SUIT
25. A BOOKCASE

